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Institute for AI and
Autonomous Systems

Human-Centered Interaction and Collaboration in Robotics

Social Event Recommendation Agent

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Project Description

In your previous assignments, you realized several multi-modal behaviors on the robot, e.g. waving and greeting a person, nodding, swirling, etc. You also created a conversational assistant that provides information about the MAS admission procedure.

In this project, you will build a socially interactive robot that acts as a social event recommendation agent to help users in finding a list of suitable social event options as per their preferences. For this, you will use the depicted form of the robot in the qiBullet simulation environment and provide it with social interaction capabilities as follows:

1. The robot detects a face from the video stream provided by your webcam. Optionally, it can verify whether the face belongs to its authorized users (i.e. your group members).
2. When a face is detected, the robot initiates a multi-modal greeting behavior. Optionally, you can personalize the greeting based on the identity of the person.
3. The robot is aware that there can be noise and errors in face detection and verification which can lead to temporary loss or misclassification of the face. The robot possesses necessary checks to deal with such scenarios.
4. After greeting, the robot initiates a conversation to assist the user in finding a preferred list of social event options.
5. The robot has an internal model of event alternatives suiting different preferences and uses this model to infer which event choice will best satisfy a person's preference. Optionally, the robot can also give a ranking of facilities that match the preference of the person. (*Hint: Use a Bayesian network*).
6. During the conversation, the robot shows suitable illustrative gestures to accompany speech.
7. After the conversation is complete, the robot bids farewell in a socially appropriate manner.
8. The robot is capable of filtering abusive language and refraining politely from engaging in such exchanges.
9. Optionally, the robot accepts speech input.
10. Name your agent and create a marketing slogan for it.

Project Timeline

- **07.05.2026:** *Project start*
- **11.06.2026:** *Checkpoint 1:* Present a short proposal (1–2 pages) that includes the design of your Bayesian network for preference-based event selection.
- **25.06.2026:** *Checkpoint 2:* Present the design of your software architecture and upload a video showing individual components in action (e.g. face detection, greeting, simple dialog flow with gestures, farewell, Bayesian model).
- **16.07.2026:** *Final project demonstration:* A working prototype, max. 5-slide presentation explaining technical design and challenges. Each group gets max. 20 minutes (including set-up, demonstration, presentation, Q&A).

Evaluation Criteria

- Number of demonstrated multi-modal social interaction capabilities
- Project proposal
- Bayesian network design
- Dialog agent design
- Visualization of the software architecture
- Object-oriented design
- Documentation of code
- Performance in Q&A

Submission

- Submit the requested documents for each project milestone in LEA.
- Late submissions will not be accepted.